



OPEN ACE2 and TMPRSS2 gene expression is reduced acutely in SARS-CoV-2 patients but returns to normal with recovery

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During the SARS-CoV-2 pandemic, angiotensin-converting enzyme 2 (*ACE2*) was identified as the major entry receptor for the virus and transmembrane serine protease 2 (*TMPRSS2*) as an important SARS-CoV-2 entry factor. Previous studies investigating the impact of *ACE2* and *TMPRSS2* gene expression on SARS-CoV-2 susceptibility in adults have yielded inconsistent results, thereby underscoring the need for further research in this domain. We obtained nasopharyngeal swabs from infected adults during the acute and late convalescent phase of SARS-CoV-2 infection and compared the expression of both genes with non-infected household member contacts. We found that *ACE2* and *TMPRSS2* gene expression is temporarily reduced during the acute phase of SARS-CoV-2 infection presumably due to viral disruption of transcription. Post-recovery, however, the expression of *ACE2* and *TMPRSS2* was comparable to non-infected household contacts. The lack of significant differences in *ACE2* and *TMPRSS2* gene expression between SARS-CoV-2-positive adults and uninfected household controls suggests that factors influencing susceptibility to SARS-CoV-2 infection in adults may extend beyond *ACE2* and *TMPRSS2* expression. However, the findings should be interpreted with caution due to the study's limited sample size and the heterogeneity of COVID-19 cases. Other non-physiological factors, such as enhanced hygiene practices following the infection of a household member, may also contribute to absence of infection among healthy controls.

Keywords SARS CoV-2, *ACE2*, *TMPRSS2*

The global pandemic of Coronavirus disease 2019 (COVID-19) caused by the SARS-CoV-2 virus was a major global health challenge. Angiotensin-converting enzyme 2 (*ACE2*) has been identified as the major entry receptor for the virus and transmembrane serine protease 2 (*TMPRSS2*) as an important SARS-CoV-2 entry factor¹.

Co-expression of *ACE2* and *TMPRSS2* genes has been detected by single-cell RNA-sequencing analyses on goblet secretory cells in the nasal mucosa, type-2 pneumocytes in the lungs and absorptive enterocytes in the small intestine². As the respiratory and olfactory epithelium in the nasal mucosa is one of the first contact points for the virus, the expression of these genes in nasal epithelium was thought to affect SARS-CoV-2 susceptibility and transmissibility.

Pre-pandemic data indicate that children exhibit reduced expression of *ACE2* in their nasal and bronchial epithelium when compared to adults, possibly accounting for the reduced SARS-CoV-2 susceptibility among children³. Moreover, children with lower levels of *ACE2* and *TMPRSS2* expression were more likely to remain

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SARS-CoV-2 negative despite being exposed to a SARS-CoV-2 positive adult household contact in one Canadian study⁴, although another group found contrasting results measuring transmembrane ACE2 and TMPRSS2⁵.

Literature on the post-recovery expression levels of *ACE2* and *TMPRSS2* following acute SARS-CoV-2 infection remains limited. To our knowledge, only one recent study by Grisard et al. has assessed the changes in *ACE2* and *TMPRSS2* expression levels in asymptomatic and symptomatic SARS-CoV-2 infected adults before, during and after infection. Their study reported that symptomatic patients exhibited significantly lower *ACE2* expression levels both before and after infection compared to asymptomatic individuals⁶. In contrast, *TMPRSS2* expression levels were initially high prior to infection, decreased during the acute phase, but did not return to pre-infection levels. Our study seeks to further investigate this relationship. Moreover, few studies have examined the relationship between *ACE2*, *TMPRSS2* expression and infection status between SARS-CoV-2 infected adults and their non-infected household member contacts. In this study, we aimed to describe the *ACE2* and *TMPRSS2* expression levels in nasopharyngeal swab (NPS) samples collected from SARS-CoV-2 positive adults during the acute and late convalescent phases of infection, and compared these findings to those of uninfected household contacts.

Methodology

NPS samples were obtained from adults who had tested positive by PCR testing for SARS-CoV-2 during the acute phase and the late convalescent phase of their infection, with each pair of samples collected from the same individual. Acute phase NPS samples were collected at the time of their hospital admission between February to May 2020. Late convalescent NPS samples were taken between June to September 2021, well after recovery from the primary infection. NPS collection from uninfected household members was performed on the same day as when late convalescence NPS were taken from patients infected with SARS-CoV-2.

At the start of the study, all patients who tested positive for SARS-CoV-2 infection in Singapore regardless of severity were hospitalised for isolation and observation and swabs were obtained from their household members for contact tracing. Household members were defined as individuals residing in the same household as the infected patient, thereby having had close contact and relatively consistent exposure to the infected individual. Available anonymized and deidentified samples from both infected cases who had a repeat sample available, as well as their household contacts who tested negative for SARS-CoV-2 infection by RT-PCR and had swabs available, were obtained from the clinical molecular diagnostic laboratory. This study was approved by the Domain Specific Institutional Review Board (National Healthcare Group Domain Specific Review Board, 2020/00120) and performed in accordance with the relevant guidelines and regulations, with a waiver of consent granted as the samples were irreversibly anonymized to the study team.

All NPS were placed immediately into viral transport media (220526, BD Universal Viral Transport Media, Becton, Dickinson and Company, Franklin Lakes, NJ). A total of 19 paired samples and 12 healthy control samples were collected from participants at the National University Hospital, Singapore.

At the time, all samples were required to demonstrate viral inactivation before removal from the Biosafety level 3 (BSL-3) facility. Viral inactivation of VTM was performed with Trizol reagent (10% v/v) for 30 min at room temperature in the BSL-3 prior to removal from the facility. RNA was extracted with the Direct-zol RNA Miniprep kit (R2050, Zymo Research, Irvine, CA). Additional Trizol reagent was added to the samples to make up a 1:1 sample to Trizol ratio. An equal volume of absolute ethanol was added to the sample-Trizol mixture. For the same sample ID, the mixture was loaded to the same Zymo-Spin™ IICR column and centrifuged at 10,000 $\times g$ for 30s multiple times (800 μ l per spin) until the total volume was passed through the column. Subsequent steps were performed according to the manufacturer's protocol. RNA was finally eluted in 40 μ l of nuclease-free water and stored at -80 °C. Due to the relatively low concentration of RNA in these specimens (0.5–1 ng/ μ l), normalizing to the same RNA concentration for each sample was challenging. Consequently, the same volume of extracted RNA was used for reverse transcription, and normalisation was subsequently performed to the reference 18S gene.

Total RNA was reverse-transcribed using the High Capacity cDNA Reverse Transcription Kit (4368814, Applied Biosystems, Thermo Fisher Scientific, Waltham, MA), on a Bio-Rad T100 Thermal Cycler, followed by qRT-PCR using the following primers:

Target	Forward (5'→3')	Reverse primer (5'→3')
ACE2	AGTCCATATGCAGACCAAGCA	TTAGCCACTCGCACATCCTC
TMPRSS2	TGGAAGTTCATGGGAGCAA	AAGTTTGGTCCGTAGAGGCG
18S	GACACGGACAGGATTGACAGATTG	TGCCAGAGTCTCGTTCGTTATCG

The primers were designed through the NCBI primer design tool, using conventional parameters that are appropriate for primer selection, such as primer length (~19–22 bases); Tm approximately 60; GC content (45–55%); low self-complementarity. The melting curves of each primer revealed a single major curve indicating that a single PCR product was amplified (Supplementary Fig. 1). Negative controls with only nuclease-free water were also measured in place of the sample cDNA, which showed no gene expression. Amplification was performed for 45 cycles on 384 well plates in the QuantStudio 5 Real-Time PCR system (Applied Biosystems, Thermo Fisher Scientific, Waltham, MA), using PerfeCTa SYBR® Green FastMix with Low ROX (Quantabio, Beverly, MA). The settings for reverse-transcription and qPCR are included in the supplementary material.

All samples were measured in duplicates. Results were normalized to the Ct values of the respective sample's 18S reference gene. The target gene expression was calculated by the $2^{-\Delta\Delta C_t}$ method and presented as fold-change, relative to the average ΔC_t value of uninfected household contacts. Each subject's target gene expression

	SARS-CoV-2 patients	Uninfected contacts	<i>p</i> -value
Sample size	19	12	
Age, mean $\pm$ SD (range)	49.5 $\pm$ 14.5 (27–77)	40.8 $\pm$ 9.7 (29–56)	0.076 ¹
Gender, n(%)			
Female	9 (47.4)	8 (66.7)	0.461 ²
Male	10 (52.6)	4 (33.3)	

Table 1. Demographics of study cohort. ¹2-tailed independent t-test. ²Fisher exact test.

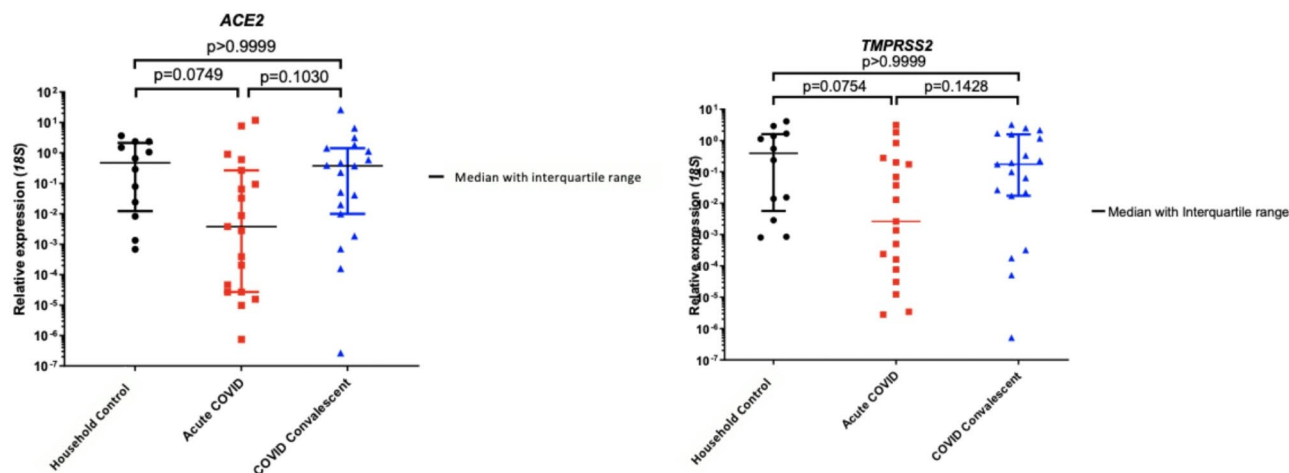


Fig. 1. Gene expression of ACE2 and TMPRSS2 in the study cohort.

was calculated relative to their own 18S expression within the same sample and subsequently compared across the groups.

Statistical analysis was performed using SPSS Statistics (version 25.0, SPSS Inc., Chicago, IL) and the GraphPad Prism 8 Software. Characteristics of SARS-CoV-2 positive patients and healthy household members were compared using either 2-tailed independent t-test for continuous variables or Fischer's exact test for categorical variables. Gene expression levels are described as median $\pm$ interquartile range (IQR). Given the data was non-parametric, Kruskal-Wallis test was used to determine the statistical significance of differences in median *ACE2* and *TMPRSS2* expression between the household member controls and acute SARS-CoV-2 positive patients, as well as between the acute and convalescent groups. A *p*-value of less than 0.05 was considered significant. In this study, the *ACE2* and *TMPRSS2* measurements refer to total *ACE2* and *TMPRSS2* levels.

Results

Demographics

The baseline demographics of the study population are described in Table 1. Most cases were mild, with only six (33%) requiring oxygen therapy at any time during their acute infection. The median time between NPS collection during the acute phase and the late convalescent phase was 522 days (IQR 509–564 days). The time period between collection of uninfected household members' swab sample and that of an infected household member's acute swab sample reflects the time period that uninfected household members remained negative for SARS-CoV-2, and this assumption was made given the extent of contact tracing done by Singapore's public health authorities at the time of the study. Rigorous contact tracing and stringent testing of both symptomatic and asymptomatic individuals was performed in Singapore during this period. NPS collection from uninfected household members was performed on the same day as when late convalescence NPS were taken from patients infected with SARS-CoV-2.

ACE2 and TMPRSS2 gene expression in SARS-CoV-2 positive patients

Patients infected with SARS-CoV-2 had lower *ACE2* expression during the acute phase compared to uninfected household members (median relative expression 0.004 (7.46×10^{-7} –11.747) vs. 0.472 (0.001–3.66), *p*=0.0749, respectively; Fig. 1). Similarly, *TMPRSS2* expression was also reduced during the acute phase of infection, compared to uninfected household members (median relative expression 0.003 (2.81×10^{-6} –3.11) vs. 0.39 (0.001–4.09), *p*=0.0754).

Patients infected with SARS-CoV-2 had lower *ACE2* expression during the acute phase compared to the late convalescent phase, though this did not reach statistical significance (median relative expression 0.004 (7.46×10^{-7} –11.747) vs. 0.374 (2.6×10^{-7} –26.09), *p*=0.103, respectively). A similar trend was observed for

TMPRSS2, with expression levels increasing during the late convalescent phase (median relative expression 0.175 (5.13×10^{-7} –3.20) vs. 0.003 (2.81×10^{-6} –3.11), $p = 0.143$).

Critically, when compared to uninfected household members after recovery from the infection, there was no significant difference in gene expression of both *ACE2* and *TMPRSS2* for SARS-CoV-2 positive patients during the late convalescence phase (median *ACE2* relative expression 0.374 (2.6×10^{-7} –26.09) vs. 0.472 (0.001–3.66), $p > 0.999$; median *TMPRSS2* relative expression 0.175 (5.13×10^{-7} –3.20) vs. 0.39 (0.001–4.09), $p > 0.999$).

Discussion

The reasons for differential susceptibility to SARS-CoV-2 infection remain unclear. Genome-wide association studies have not yielded conclusive results and data on cross protection from seasonal coronaviruses have not provided answers⁷. Attention has been focused on *ACE2* and *TMPRSS2* activity. Since the lower expression of nasopharyngeal *ACE2* and possibly also *TMPRSS2* has been suggested as a protective mechanism in children against SARS-CoV-2 infection, we also hypothesized that non-infected household member contacts may have lower *ACE2* and *TMPRSS2* expression compared to infected patients at their baseline. However, our study revealed that *ACE2* and *TMPRSS2* transcription levels were actually similar in infected patients in the convalescent (presumably baseline) phase compared to their uninfected household member contacts. This is despite observing significant differences between gene expression levels during acute infection compared to uninfected controls.

ACE2 has been well characterized to be the main cell surface entry receptor for SARS-CoV-2, and *TMPRSS2* is known as an important priming enzyme facilitating the virus's fusion on the cell membrane to gain entry into the host cells⁸. Similar to Nikiforuk et al., who demonstrated a decrease in both the soluble and transmembrane isoforms of nasopharyngeal *ACE2* and *TMPRSS2*⁴, we also observed a decrease in the expression of these genes in SARS-CoV-2 positive patients compared to uninfected controls during the acute phase. The reduction may arise either as a host response to prevent viral dissemination by downregulating *ACE2* and *TMPRSS2* expression, or conversely, depletion to facilitate the viral entry and replication. For the case of *ACE2*, it has been shown that the SARS-CoV-2 S-protein is responsible for the *ACE2* downregulation⁹. All variants of SARS-CoV-2 S proteins have been shown to retain the ability to do so in vitro, albeit to a lesser extent in Omicron variants¹⁰. SARS-CoV-2's disruption of *ACE2* expression then in turn has been associated with adverse physiological effects on the host, through the dysregulation of the local renin angiotensin system, potentially contributing to SARS-CoV-2's cardiovascular symptoms¹¹. Taken together, the ability of SARS-CoV-2 S protein from all variants being able to down-regulate *ACE2* expression, may contribute to continual observation of these adverse physiological effects clinically¹². Thus, the restoration of the *ACE2* expression following infection may be vital for complete post infection recovery. On the other hand, *TMPRSS2* reduction has been shown in both SARS-CoV and SARS-CoV-2¹³, in which the reduction affected leukocyte infiltration which may contribute to the SARS-CoV and SARS-Cov-2 pathologies. Therefore, it is more plausible that the reduction of *ACE2* and *TMPRSS2* are virally induced to facilitate its infection, replication and pathology.

Although we have shown that alterations in *ACE2* and *TMPRSS2* expression may not be that common in acute SARS-CoV-2 infection in seminal fluid, for example¹⁴; others have shown persistently elevated *ACE2* activity in plasma¹⁵. However, the longitudinal expression of *ACE2* in the nasal cavity of patients with SARS-CoV-2 has not been well described previously.

In this study, we found that *ACE2* and *TMPRSS2* transcription levels were temporarily reduced during the acute infection phase in infected patients, but recovered during the convalescent phase to levels comparable to household member controls who are not-infected.

In comparison, a study by Grisard et al. reported significantly higher *TMPRSS2* expression levels in symptomatic patients before infection compared to asymptomatic patients, though there was no difference during the convalescent phase⁶. In addition, *ACE2* expression levels were significantly lower in symptomatic patients pre and post-infection. These findings highlight the heterogeneity in *ACE2* and *TMPRSS2* expression levels and suggest that there remain undiscovered factors beyond the expression of *ACE2* and *TMPRSS2* that can affect the susceptibility to SARS-CoV-2 infection in adults.

Limitations of this study include the relatively small sample size and the low RNA concentration obtained from NPS samples after viral inactivation. This limited our ability to distinguish between the transmembrane and soluble isoforms of *ACE2*. Further studies should include assays designed to differentiate between the isoforms of *ACE2*. As the samples were anonymized and deidentified, we did not have data on vaccination status of infected patients which might have affected their convalescent samples. We also did not have data on concomitant medications including ACE-inhibitors or ACE-receptor blockers. Household contacts were only sampled once during the study and there was lack of objective serological evidence showing that they remained SARS-CoV-2 negative during the immediate post-exposure period; we assume that household contacts remained negative from the date when the acute swab sample was taken and the date when uninfected household contacts' swab samples were collected, given the extensive contact tracing that was performed in Singapore during the study period. The lack of infection in the household contacts may also be confounded by non-physiological factors, such as vaccination (although this was only rolled out late in the study period) and the heightened state of hygiene practices following confirmation of infection of a household member.

Conclusion

Our study adds to the literature on the role of *ACE2* and *TMPRSS2* gene expression levels and their changes in the nasopharynx during the acute and convalescent phase of SARS-CoV-2 infection. These findings suggest that while *ACE2* and *TMPRSS2* expression were disrupted during acute phases of infection, their disruption is temporary, with recovery of the expression occurring during recovery from SARS-CoV-2 infection. With levels

of *ACE2* and *TMPRSS2* expression being comparable between the convalescent phase infected patients and the non-infected controls, the levels of expression alone may not fully explain the differential susceptibility to SARS-CoV-2 infection. Further research with a larger sample size and concurrent protein validation should be considered, as well as to investigate other host traits that may influence susceptibility to SARS-CoV-2 infection.

Data availability

The datasets generated and analysed during the study are available in the Figshare repository, <https://figshare.com/s/ebbcff6203da9be90aaf?file=51730850> (DOI: <https://doi.org/10.6084/m9.figshare.28226699>).

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References

- Vieira, C. et al. Downregulation of Membrane-bound angiotensin converting enzyme 2 (ACE2) receptor has a pivotal role in COVID-19 immunopathology. *Curr. Drug Targets* **22**, 254–281 (2021).
- Ziegler, C. et al. SARS-CoV-2 receptor ACE2 is an interferon-stimulated gene in human airway epithelial cells and is detected in specific cell subsets across tissues. *Cell* **181**, 1016–1035 (2020). e1019.
- Bunyavanich, S., Do, A. & Vicencio, A. Nasal gene expression of Angiotensin-Converting enzyme 2 in children and adults. *JAMA* **323**, 2427–2429 (2020).
- Nikiforuk, A. et al. The contrasting role of nasopharyngeal angiotensin converting enzyme 2 (ACE2) transcription in SARS-CoV-2 infection: A cross-sectional study of people tested for COVID-19 in British Columbia, Canada. *EBioMedicine* **66**, 103316 (2021).
- Hasan, M. R. et al. Nasopharyngeal expression of Angiotensin-Converting enzyme 2 and transmembrane Serine protease 2 in children within SARS-CoV-2-Infected family clusters. *Microbiol. Spectr.* **9**, e00783–e00721 (2021).
- Grisard, H. B. et al. ACE2 and TMPRSS2 expression in patients before, during, and after SARS-CoV-2 infection. *Front. Cell. Infect. Microbiol.* **14**. <https://doi.org/10.3389/fcimb.2024.1355809> (2024).
- Wong, L. et al. Early seasonal coronavirus seroconversion did not produce cross-protective SARS-CoV-2 antibodies. *J. Infect.* **86**, e10–e12 (2023).
- Rossi, A. D. et al. Association between ACE2 and TMPRSS2 nasopharyngeal expression and COVID-19 respiratory distress. *Sci. Rep.* **11**, 9658 (2021).
- Lu, Y. et al. SARS-CoV-2 down-regulates ACE2 through lysosomal degradation. *Mol. Biol. Cell.* **33**. <https://doi.org/10.1091/mbc.E22-02-0045> (2022).
- Maeda, Y. et al. Differential ability of Spike protein of SARS-CoV-2 variants to downregulate ACE2. *Int. J. Mol. Sci.* **25**. <https://doi.org/10.3390/ijms25021353> (2024).
- Viveiros, A. et al. SARS-CoV-2 infection downregulates myocardial ACE2 and potentiates cardiac inflammation in humans and hamsters. *Am. J. Physiol. Heart Circ. Physiol.* **323**, H1262–H1269 (2022).
- Vishwakarma, N., Goud, R. B., Tirupattur, M. P. & Katwa, L. C. The Eye of the Storm: Investigating the Long-Term Cardiovascular Effects of COVID-19 and Variants. *Cells* **12**. <https://doi.org/10.3390/cells12172154> (2023).
- Liu, X. et al. Decreased TMPRSS2 expression by SARS-CoV-2 predicts the poor prognosis of lung cancer patients through metabolic pathways and immune infiltration. *Aging (Albany NY)* **14**, 73–108. <https://doi.org/10.18632/aging.203823> (2022).
- Huang, Z. et al. Effects of acute severe acute respiratory syndrome coronavirus 2 infection on male hormone profile, ACE2 and TMPRSS2 expression, and potential for transmission of severe acute respiratory syndrome coronavirus 2 in semen of Asian men. *F S Sci.* **3**, 29–34 (2022).
- Patel, S. K. et al. Plasma ACE2 activity is persistently elevated following SARS-CoV-2 infection: implications for COVID-19 pathogenesis and consequences. *Eur. Respir J.* **57**, 2003730. <https://doi.org/10.1183/13993003> (2021).

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Author contributions

V.H.E.C: analysis, manuscript writing. J.K.T: analysis, manuscript writing. R.G: conduct, analysis, manuscript writing. S.N: conduct, analysis, manuscript revision. D.J.T: conduct, analysis, manuscript revision. K.S.T: conduct, analysis, manuscript writing. R.S.Y.F: design, conduct, analysis, manuscript revision. P.A.T: design, conduct, analysis, manuscript writing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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